

name: Asmaa Ahmed Maryah

ID : 811634546

major: IS

Tasks

Difference Between Data Scientist, ML Engineer, Data Engineer, and Data Analyst

1. Data Scientist

A Data Scientist focuses on analyzing large amounts of data to discover patterns, trends, and useful insights. They use statistics, machine learning, and programming to solve business problems and make predictions.

Main Responsibilities:

- * Building predictive models
- * Data cleaning and preprocessing
- * Statistical analysis
- * Machine learning model development
- * Business problem solving

Required Skills:

- * Python / R
- * SQL
- * Machine Learning
- * Statistics
- * Data Visualization
- * Problem Solving

2. Machine Learning Engineer

A Machine Learning Engineer focuses on designing, building, and deploying machine learning models into real-world applications. Their work is more engineering-focused than analysis-

focused.

Main Responsibilities:

- * Deploying ML models into production
- * Optimizing model performance
- * Building ML pipelines
- * Model monitoring and maintenance
- * Working with software systems

Required Skills:

- * Python
- * Machine Learning
- * Deep Learning
- * MLOps
- * APIs
- * Cloud Platforms
- * Software Engineering

3. Data Engineer

A Data Engineer focuses on building and managing the systems that collect, store, and process data. They make sure data is available, clean, and accessible for analysts and scientists.

Main Responsibilities:

- * Building data pipelines
- * Database management
- * Data warehousing
- * ETL processes (Extract, Transform, Load)
- * Big Data tools management

Required Skills:

- * SQL
- * Python / Java / Scala
- * Hadoop
- * Spark
- * Data Warehousing
- * Database Systems
- * Cloud Services

4. Data Analyst

A Data Analyst focuses on interpreting data and creating reports to help businesses make decisions. They work mainly with dashboards, reports, and business insights.

Main Responsibilities:

- * Data reporting
- * Dashboard creation
- * Business analysis
- * Data visualization
- * KPI tracking

Required Skills:

- * SQL
- * Excel
- * Power BI / Tableau
- * Statistics basics
- * Data Visualization
- * Business Understanding

Simple Comparison Table

Role	Main Focus	Main Goal
Data Scientist	Analysis + Prediction	Solve complex business problems
ML Engineer	Model Deployment	Put ML models into production
Data Engineer	Data Infrastructure	Build data systems and pipelines
Data Analyst	Reporting + Visualization	Support business decisions

2. Examples of Big Data Characteristics (5 Vs)

1. Volume

Volume means the huge amount of data generated every day.

Example:

Social media platforms like Facebook and Instagram generate billions of posts, photos, videos, and messages daily.

2. Velocity

Velocity means the speed at which data is generated and processed.

Example:

Online payment systems like PayPal process thousands of transactions every second and need real-time analysis for fraud detection.

3. Variety

Variety means different types of data such as text, images, videos, audio, and sensor data.

Example:

A hospital stores patient records as text files, X-ray images, MRI scans, voice reports, and lab test results.

4. Veracity

Veracity means the quality, accuracy, and trustworthiness of data.

Example:

Customer reviews on e-commerce websites may contain fake reviews, incomplete information, or incorrect ratings, which affect decision-making.

5. Value

Value means turning raw data into useful business insights.

Example:

Amazon analyzes customer purchase history to recommend products and improve sales.

3. Messy Data vs Noisy Data

What is Messy Data?

Messy Data means data that is poorly organized, incomplete, duplicated, inconsistent, or stored in the wrong format. It makes analysis difficult and can lead to incorrect results.

Examples of Messy Data

1. Missing Values

2. Duplicate Records

Example:

The same customer appears twice in the database.

3. Inconsistent Format

Example:

Date stored as:

* 12/04/2026

* April 12, 2026

* 2026-04-12

Different formats create confusion.

4. Wrong Data Type

Example:

Phone number stored as a number instead of text.

How to Handle Messy Data

- * Remove duplicate records
- * Fill missing values using mean, median, mode, or prediction
- * Standardize formats (dates, names, units)
- * Correct wrong data types
- * Remove invalid records
- * Use data cleaning tools like Python Pandas, SQL, or Excel

What is Noisy Data?

Noisy Data means data that contains random errors, unusual values, or incorrect values caused by human mistakes, device errors, or measurement problems.

It affects model accuracy because the values themselves are incorrect.

Examples of Noisy Data

1. Wrong Age Value

Age = 250 is unrealistic and considered noise.

2. Sensor Error

Example:

A temperature sensor shows 300°C in a normal office.

3. Typing Mistake

Example:

Salary entered as 50000 instead of 5000.

How to Handle Noisy Data

- * Detect outliers using statistics
- * Use smoothing techniques
- * Binning method
- * Regression methods
- * Clustering methods
- * Remove extreme abnormal values
- * Validate data ranges

4. Handling Spelling Errors in Data

When the same country, city, or name is written in different ways, this is considered a data quality problem and usually falls under Messy Data because the format is inconsistent.

Example

The same country may appear like this:

- * Egypt
- * egypt
- * EGYPT
- * Egypet
- * Eygpt

All of them should represent only one value: Egypt

This causes problems in analysis because the system may treat them as different countries.

How to Solve It

1. Standardization

Convert all text into one standard format.

Example:

- * Make all letters lowercase or uppercase
- * Remove extra spaces
- * Use one official spelling only

Example:

egypt → Egypt

EGYPT → Egypt

2. Data Validation Rules

Create rules during data entry.

Example:

Instead of typing manually, use a dropdown list of country names so users choose from predefined values.

This reduces spelling mistakes.

3. Fuzzy Matching

Use algorithms that detect similar words even if spelling is wrong.

Example:

Egypt → Egypt

Unted States → United States

Common techniques:

- * Levenshtein Distance
- * Similarity Matching
- * Python libraries like fuzzywuzzy or rapidfuzz

4. Reference Table

Create a mapping table.

The system automatically replaces wrong values.

Detecting and Handling Outliers

What are Outliers?

5. Outliers are data values that are very far from the normal range of the dataset.

They are unusual values that may happen because of human mistakes, sensor errors, or rare real events.

How to Detect Outliers

1. Using Visualization

Graphs help us easily notice unusual values.

Examples:

- * Box Plot

- * Scatter Plot

- * Histogram

2. Using Z-Score Method

This method checks how far a value is from the mean.

3. Using IQR Method (Interquartile Range)

This is one of the most common methods.

Formula:

$$IQR = Q3 - Q1$$

Then calculate:

$$\text{Lower Bound} = Q1 - 1.5 \times IQR$$

$$\text{Upper Bound} = Q3 + 1.5 \times IQR$$

Any value outside this range is considered an outlier.

How to Handle Outliers

1. Remove the Outlier

If the value is clearly wrong.

Example:

Age = 300 years

This should be removed.

2. Replace the Value

Replace it using:

- * Mean

- * Median

- * Mode

3. Capping (Winsorization)

4. Keep It

Sometimes the outlier is real and important.

Example:

A millionaire's salary in salary analysis

In this case, we should keep it.

Difference Between Error and Real Outlier

Examples of the Four Data Integration Methods

What is Data Integration?

Data Integration means combining data from different sources into one unified view so the data can be used easily for analysis and decision-making.

1. Manual Data Integration

This means people manually collect data from different sources and combine it by themselves.

Problem

- * Slow
- * High chance of human error
- * Not suitable for large data

2. Middleware Data Integration

This uses software that acts as a bridge between different systems and helps them communicate and exchange data.

Benefit

- * Real-time connection
- * Less manual work

3. Application-Based Integration

This means one application is responsible for collecting and integrating data from different systems.

Example

An ERP system like SAP collects data from accounting, HR, inventory, and sales departments into one application.

Benefit

- * Centralized management
- * Easy access

4. Uniform Access Integration

This method creates one virtual interface where users can access data from multiple sources without moving the actual data.

Example

A business intelligence tool allows managers to query data from multiple databases using one dashboard without physically merging the databases.

Benefit

- * Fast access
- * No need to duplicate data

t-Distribution & F-Distribution

1. t-Distribution

What is it?

The t-distribution is a probability distribution used when:

- * Sample size is small (usually $n < 30$)
- * Population standard deviation is unknown

It is similar to the normal distribution but has heavier tails (more spread).

When to Use t-Distribution

- * Small datasets
- * Unknown population standard deviation
- * Hypothesis testing for means

Example

You want to test if students' average score is 70, but you only have a sample of 10 students.

You use t-distribution instead of normal distribution.

2. F-Distribution

What is it?

The F-distribution is used to compare two variances and is mostly used in:

- * ANOVA (Analysis of Variance)
- * Model comparison in machine learning

It is always right-skewed (not symmetric).

When to Use F-Distribution

- * Comparing two or more group variances
- * Testing if multiple means are equal (ANOVA)
- * Regression model significance

Example

You want to check if two teaching methods give different score variations.

You compare:

- * Variance of group A students
- * Variance of group B students

Fields Where Hypothesis Testing is Used

What is Hypothesis Testing?

Hypothesis testing is a statistical method used to make decisions about a population based on sample data. It helps answer questions like: *Is this difference real or just by chance?*

1. Data Science & Machine Learning

Hypothesis testing is used to validate models and compare results.

Examples:

- * Comparing two ML models' accuracy

- * Checking if a feature improves performance

- * A/B testing for model versions

2. Business & Marketing

Companies use it to make data-driven decisions.

Examples:

- * Testing if a new marketing campaign increases sales

- * Comparing two advertisement designs (A/B testing)

- * Checking if price change affects demand

Example: Amazon tests different homepage layouts to see which increases purchases.

3. Healthcare & Medicine

Used to test treatment effectiveness.

Examples:

- * Does a new drug reduce blood pressure more than the old one?

- * Comparing recovery rates between two treatments

- * Clinical trials

Hospitals and pharmaceutical companies rely heavily on hypothesis testing.

4. Education

Used to evaluate teaching methods and student performance.

Examples:

- * Does a new teaching method improve exam scores?

- * Comparing online vs traditional learning outcomes

- * Evaluating student improvement after training programs

5. Finance & Banking

Used for risk analysis and decision-making.

Examples:

- * Testing if a new investment strategy increases profit

- * Checking if loan approval models are accurate

- * Fraud detection validation

Banks like HSBC use it for risk modeling.

6. Social Sciences

Used in psychology, sociology, and human behavior studies.

Examples:

- * Does social media affect attention span?

- * Comparing behavior between groups of people

- * Testing survey assumptions

7. Engineering & Quality Control

Used in manufacturing and production systems.

Examples:

- * Testing if machine A produces fewer defects than machine B

- * Quality control in factories

- * Checking production consistency

Simple Summary

Hypothesis testing is used anywhere we need to make decisions from data:

- * Business → marketing & sales

- * Healthcare → drugs & treatments

- * ML/Data Science → model validation

- * Education → teaching evaluation

- * Finance → risk & fraud

- * Engineering → quality control